# MATH 101 Unit 4: Exponentials, Logarithms, and Systems - Practice 1

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Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

**1.** Evaluate exactly using logarithmic properties; show the property used. Here log means base 10.

(*a*) $\log(0.2) + \log(500)$

(*b*) $\dfrac{\log_3(8)}{\log_3(4)}$

**2.** Solve both equations exactly. Check the logarithmic domain in (b).

(*a*) $4^{x-1} = 2^{x+2}$

(*b*) $\log_2(x-1) + \log_2(3) = 2$

**3.** Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain.

$f(x) = \log_2(x-1) + 1$

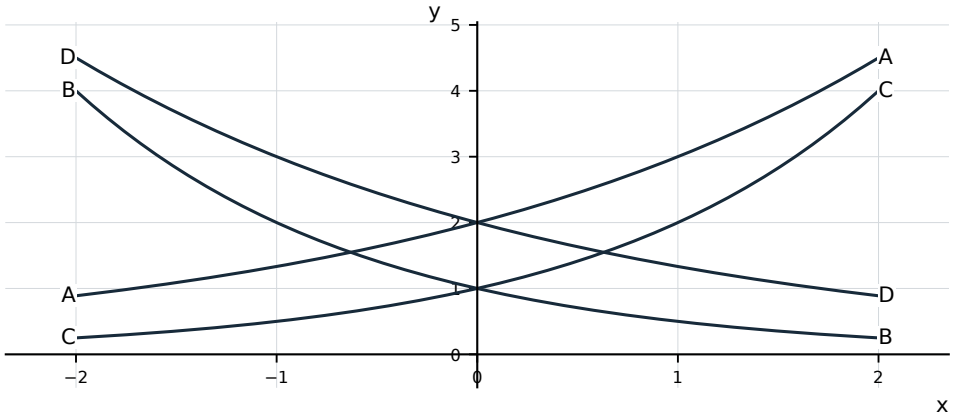
**4.** Match each function to graph A, B, C, or D. Each letter is used once.

(*a*) $1(2)^x$

(*b*) $2(\frac{3}{2})^x$

(*c*) $2(\frac{2}{3})^x$

(*d*) $1(\frac{1}{2})^x$



**5.** (a) Solve the linear system. (b) Solve the nonlinear system algebraically. (c) Test whether (0,0) belongs to the region given by the inequalities; show both comparisons.

$(a)\quad x+y=3$

$x-y=1$

$(b)\quad x^2+y^2=5$

$y=x+1$

$(c)\quad x^2+y^2\le 5,\ y>x+1$

# Student answer sheet

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**Question 1**

(a) 10 pts

(b) 10 pts

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**Question 2**

(a) 8 pts

(b), with domain 12 pts

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**Question 3**

(a) Domain 4 pts

(a) Range 4 pts

(b) Asymptote 3 pts

(b) x-intercept 3 pts

(c) Left end 3 pts

(c) Right end 3 pts

# Student answer sheet

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**Question 4**

(a)

5 pts

(b)

5 pts

(c)

5 pts

(d)

5 pts

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**Question 5**

(a) Ordered pair

6 pts

(b) Intersections

10 pts

(c) Membership

4 pts

# Answer key

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**Question 1**

| | | |
|---|---|---|
| (a) | $2$ | 10 pts |
| (b) | $\frac{3}{2}$ | 10 pts |

**Question 2**

| | | |
|---|---|---|
| (a) | $x = 4$ | 8 pts |
| (b), with domain | $x = \frac{7}{3};\ x > 1$ | 12 pts |

**Question 3**

| | | |
|---|---|---|
| (a) Domain | $(1, \infty)$ | 4 pts |
| (a) Range | $(-\infty, \infty)$ | 4 pts |
| (b) Asymptote | $x = 1$ | 3 pts |
| (b) x-intercept | $(\frac{3}{2},\ 0)$ | 3 pts |
| (c) Left end | $x \to 1^+ :\ f(x) \to -\infty$ | 3 pts |
| (c) Right end | $x \to \infty :\ f(x) \to \infty$ | 3 pts |

# Answer key

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**Question 4**

| | | |
|---|---|---|
| (a) | C | 5 pts |
| (b) | A | 5 pts |
| (c) | D | 5 pts |
| (d) | B | 5 pts |

**Question 5**

| | | |
|---|---|---|
| (a) Ordered pair | $(2,\ 1)$ | 6 pts |
| (b) Intersections | $(1,\ 2);\ (-2,\ -1)$ | 10 pts |
| (c) Membership | At (0,0): 0 <= 5, but 0 > 1 is false; not in the region. | 4 pts |